Tilt-Based Control Barrier Function for Target Visibility

A camera-plane QP over the body tilt via the rotational interaction matrix

Soft–Precise–Landing

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Scope. Visibility is an *attitude* constraint — the body tilt moves the field of view — so we guarantee it with a small QP (a clamp) on the desired tilt, on the *real* camera plane. The coupling between tilt and feature is the **rotational part of the IBVS interaction matrix**, L_ω, evaluated at the measured image point: it is exact, *depth-independent*, and needs no de-rotation / virtual frame. The barrier uses image features only — scale- and depth-free (no β, no optic flow, no disturbance term). Notation follows the manuscript.

Notation

Symbol	Meaning
${}^{\mathcal{C}}\hat{\boldsymbol{r}}_i,\ {}^{\mathcal{C}}\hat{\boldsymbol{r}} = (x, y)$	camera image feature points / centroid, tangent units, per axis k
$L_\omega({}^{\mathcal{C}}\hat{\boldsymbol{r}})$	rotational interaction matrix (roll/pitch columns), $\partial\, {}^{\mathcal{C}}\dot{\hat{\boldsymbol{r}}}/\partial \boldsymbol{\omega}_{\mathrm{rp}}$
$\boldsymbol{\omega}_{\mathrm{rp}} = [\omega_\phi, \omega_\theta]$	body roll/pitch rate (the tilt rate, the control)
$\boldsymbol{\theta}, \theta_d, \theta^\star$	body tilt per axis; desired tilt (from R_d), filtered
$\boldsymbol{d}$	exogenous (translation) feature drift, $\boldsymbol{d} = {}^{\mathcal{C}}\dot{\hat{\boldsymbol{r}}}_{\mathrm{obs}} - L_\omega \boldsymbol{\omega}_{\mathrm{rp}}^{\mathrm{curr}}$
$\mathcal{R}, f$	resolution, focal (px); $\boldsymbol{\varphi}_{\max} = \mathcal{R}/(2f)$ half-FoV (tangent), the FoV edge
$\delta, \dot{\delta}$	primary-marker half-spread (loom rate)
θ_{cap}	deliverable tilt cap (60°); τ drift look-ahead; h barrier

1 Barrier on the real camera plane

Per image axis k,

$$\boxed{h_k = \varphi_{\max,k} - |{}^{\mathcal{C}}\hat{r}_k| - \delta_k}. \tag{1}$$

$h_k \geq 0 \iff$ the marker (outermost feature point) is on the sensor. ${}^{\mathcal{C}}\hat{\boldsymbol{r}}$ is the *measured* camera image point — nothing is de-rotated.

2 Tilt→feature coupling: the rotational interaction matrix

A camera rotation moves a feature point by the rotational part of the IBVS interaction matrix, a standard and **depth-independent** object evaluated at the measured point ${}^{\mathcal{C}}\hat{\boldsymbol{r}} = (x, y)$:

$$
{}^{\mathcal{C}}\dot{\hat{\boldsymbol{r}}} = L_\omega({}^{\mathcal{C}}\hat{\boldsymbol{r}})\,\boldsymbol{\omega}_{\mathrm{rp}} + \boldsymbol{d}, \qquad L_\omega = \begin{bmatrix} xy & -(1+x^2) \\ 1+y^2 & -xy \end{bmatrix}, \tag{2}
$$

where $\boldsymbol{\omega}_{\mathrm{rp}}$ is the body roll/pitch rate (the control) and $\boldsymbol{d}$ is the exogenous (translation / target) drift. L_ω contains **no depth** — the tilt coupling needs no β and no virtual frame. The drift is measured directly by stripping the rotational term from the observed feature velocity, $\boldsymbol{d} = {}^{\mathcal{C}}\dot{\hat{\boldsymbol{r}}}_{\mathrm{obs}} - L_\omega \boldsymbol{\omega}_{\mathrm{rp}}^{\mathrm{curr}}$, so it carries only the genuine target motion, not our own tilting.

For a finite tilt change the exact map is the rotation homography of the measured point; the first-order form ${}^{\mathcal{C}}\hat{\boldsymbol{r}}(\boldsymbol{\theta}) \approx {}^{\mathcal{C}}\hat{\boldsymbol{r}} + L_\omega\,(\boldsymbol{\theta} - \boldsymbol{\theta}_{\mathrm{curr}})$ is the local linearization and replaces the earlier (incorrect) ${}^{\mathcal{C}}\hat{\boldsymbol{r}} = {}^{\hat{r}+\tan\theta}$.

3 The visibility filter: a QP over the body tilt

With $m_k = \varphi_{\max,k} - \delta_k - \tau\dot{\delta}_k$ the predicted margin,

$$
\boxed{\boldsymbol{\theta}^\star = \arg\min_{\boldsymbol{\theta}}\ \|\boldsymbol{\theta} - \boldsymbol{\theta}_d\|^2 \ \ \text{s.t.}\ \ \left|{}^{\mathcal{C}}\hat{r}_k + [L_\omega(\boldsymbol{\theta} - \boldsymbol{\theta}_{\mathrm{curr}})]_k + \tau d_k\right| \le m_k, \quad |\boldsymbol{\theta}| \le \theta_{\mathrm{cap}}} \tag{3}
$$

A small 2-variable QP (the two tilt axes), with ≤ 2 binding rows plus the cap — closed-form. L_ω is near-diagonal for a centred marker (the xy off-terms vanish), so it is nearly a per-axis clamp; the off-diagonal carries the genuine roll↔pitch cross-coupling near the FoV edge.

- **Directional** — only the outward binding row is active; inward (recovery) and tangential tilts are free. No strangling.
- **Input-aware** — the θ_{cap} box never demands a tilt the inner loop cannot hold.
- **Forward-invariant** — the look-ahead uses the predicted margin in τ s: the target drift $\boldsymbol{d}$ (left) and the marker-fill rate $\dot{\delta}$ (right) shrink it, so the box tightens early and forces the inward tilt ($\dot{h} \ge -\alpha(h)$). $\tau = 0$ gives the static clamp.

4 Two-phase δ (central-marker convention)

The designated *central* marker is the target. **Phase 1 — while it decodes,** $\delta_k = 0$: the barrier is centroid-only, keeping the target centre in frame while deliberately letting the central marker grow and overflow as the UAV closes on it (its visibility *is* the guarantee; using its large extent would fire the barrier early). **Phase 2 — on its overflow** (decode-fail, hysteresis-gated against flicker ⇒ touchdown range), δ **ramps in** over a few frames from the visible markers' $\frac{1}{2}\mathrm{ptp}_k$ and tracks them thereafter. The ramp (not a step) keeps h from dropping discontinuously at engagement, but clamp the ramp gain so δ never *lags* the measured $\frac{1}{2}$ptp — a ramp slower than the fill under-protects. Within Phase 2 the visible markers themselves leave (a mild inner sawtooth); $\dot{\delta}$ is reset across those switches.

5 Feasibility floor and hand-off

The QP (3) is **infeasible** iff the predicted margin is gone ($m_k < 0$ — the marker fills the frame within τ) or even the full $L_\omega \theta_{\text{cap}}$ tilt cannot bring ${}^{\mathcal{C}}\hat{\boldsymbol{r}}$ inside. Then visibility is physically impossible $\Rightarrow$ raise the perception-floor flag and hand off to the texture-free ring flow.

6 Assumptions (and the practical challenge each carries)

1. **L_ω's sign / axis convention is calibrated.** Its *form* is the standard interaction matrix, but the body-rate→image-axis sign (roll→which axis, which direction) depends on the camera mount. *Must-do:* command a small tilt, check $\Delta\,{}^{\mathcal{C}}\hat{\boldsymbol{r}}$ matches L_ω in sign, before trusting the barrier.

2. **The filtered tilt is realized** (SO(3) loop tracks $\boldsymbol{\theta}^\star$). *Challenge:* attitude bandwidth $\Rightarrow$ lag; θ_{cap} bounds magnitude, not delay.

3. **The drift $\boldsymbol{d}$ is the translation part** (${}^{\mathcal{C}}\dot{\hat{\boldsymbol{r}}}_{\text{obs}} - L_\omega \boldsymbol{\omega}^{\text{curr}}$) and $\dot{\delta}$ the loom. *Challenge:* finite differences $\Rightarrow$ filter + binding-axis hysteresis; the disturbance enters as *measured* motion — no separate $\boldsymbol{d}_h$ term.

4. **First-order L_ω over the cycle.** Exact is the rotation homography of ${}^{\mathcal{C}}\hat{\boldsymbol{r}}$; the $L_\omega(\boldsymbol{\theta} - \boldsymbol{\theta}_{\text{curr}})$ linearization is accurate over one control step. At $\theta_{\text{cap}} = 60^\circ$ the integral (homography) is the safe choice for the boundary $\boldsymbol{\theta}$.

5. **θ_{cap} is the true deliverable tilt** (thrust unsaturated).

7 Scale-freeness and depth-freeness

The barrier (1), the coupling (2), and the box (3) contain only the measured image point ${}^{\mathcal{C}}\hat{\boldsymbol{r}}$, δ, $\varphi_{\max} = \mathcal{R}/(2f)$, the body rate, and the tilt. L_ω **is depth-independent by construction** (no $\mathcal{V}z_{\text{t}}$, no β), so the whole formulation is manifestly scale- and depth-free. The only metric quantity upstream is the geometric tracker's R_d (a controller output).

8 Implementation (per control cycle)

1. *(once)* Calibrate L_ω's sign / axis convention (Assumption 1).

2. Measure ${}^{\mathcal{C}}\hat{\boldsymbol{r}}$ (board centroid, homography). Two-phase δ: 0 while the central marker decodes; on overflow (hysteresis) ramp in the visible markers' $\frac{1}{2}$ptp.

3. Evaluate $L_\omega({}^{\mathcal{C}}\hat{\boldsymbol{r}})$; form the drift $\boldsymbol{d} = {}^{\mathcal{C}}\dot{\hat{\boldsymbol{r}}}_{\text{obs}} - L_\omega \boldsymbol{\omega}^{\text{curr}}$ and $\dot{\delta}$ (filtered).

4. Predicted margin $m = \varphi_{\max} - \delta - \tau\dot{\delta}$; build the QP (3).

5. $\boldsymbol{\theta}_d$ from R_d (equivalently the desired lateral accel via $\tan\theta_d = {}^{\mathcal{I}}a_{d,xy}/a_z$); solve $\rightarrow \boldsymbol{\theta}^\star$ (a 2-var clamp).

6. If infeasible, raise the perception-floor flag and switch the flow source to the ring estimator; else pass $\boldsymbol{\theta}^\star$ (equivalently the clamped a_{xy}) to the SO(3) inner loop.